AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 9 · STUDENT LESSON

Dataset Cards

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Cite the source. Mark unknowns. Keep the version.

Document what a dataset contains and what its evidence cannot establish.

Keep one habit: Complete photo manifest → worked card → exact update → card review lab → independent20-photo card.

MAIN LESSON · SECTION 1

What must a reader know beyond the photo count?

> Write an evidence-linked dataset card that reveals unknowns.

Current card-writing goals

I can document a dataset's source purpose row unit and version.

I can distinguish supplied card facts from unknown fields and unsupported claims.

I can update counts coverage and limitations when a dataset changes.

Retrieve split and source reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.
- Which source IDs, split assignments and preprocessing-fit boundaries must remain traceable when describing a dataset?

End goal

By the end, I can explain **dataset card**s using a traced example and a stated **limitation**.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **dataset card** records where data came from, why it exists, its fields, collection **process**, known gaps, risks, and **prohibited use**s.



LEARNING TARGET Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Read a complete source brief before filling card fields

> Separate known facts from questions that remain open.

Current purpose and supplied source facts

A dataset card is a readable account of what a dataset contains, how it was created, what it is intended for and where evidence is missing. Give each claim a source or mark it unknown. Our current fixture is Photo practice v1: a teacher-created fictional set for learning card writing and a toy visible-spot rule. It contains18 photo rows from6 fictional plants,3views each, one tablet, indoor fluorescent light.

No real personal photos, disease diagnosis or trained-model result is supplied.

Complete eighteen-photo manifest

Plant ID	Photo IDs	Known condition
P1	P1a P1b P1c	Same tablet, indoor fluorescent
P2	P2a P2b P2c	Same tablet, indoor fluorescent
P3	P3a P3b P3c	Same tablet, indoor fluorescent
P4	P4a P4b P4c	Same tablet, indoor fluorescent
P5	P5a P5b P5c	Same tablet, indoor fluorescent
P6	P6a P6b P6c	Same tablet, indoor fluorescent

Unknown fields and honest status

The label vocabulary is visible spot/no visible spot/review, but per-photo labels and their annotation history are not supplied. The exact creation date, detailed image specification, licensing terms for any external reuse and split assignments are not provided. Do not invent

them. This teacher-authored classroom fixture is not an external dataset whose permission has been verified.

Mark these fields not supplied or not assigned, and identify the question for the teacher.

▼ Review · Familiar trip-inspection fixture

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Trip ID	Duration in minutes	What is visible
A	8	A recorded duration
B	None	An unknown duration
B	None	The same ID appears again
C	-2	A negative duration
D	12	A recorded duration

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **dataset card**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Build the card as claim, evidence and status

> The reader should be able to locate the fact or see the unanswered question.

Complete worked dataset card

Card field	Evidence-linked Photo practice v1 entry
Purpose / owner	Teacher-maintained fictional classroom card-writing and toy visible-spot practice
Source / version	Teacher-created fictional fixture, Photo practice v1; detailed creation date not supplied
Row / group / counts	One photo row; related views share plant ID;18photos/6plants/3views each
Conditions	One tablet; indoor fluorescent light; other conditions not documented
Labels	Vocabulary visible spot/no visible spot/review; per-photo labels and detailed annotation history not supplied
Changes / quality	No corrections or transformations supplied; completeness/measurement accuracy not established
Splits	Not assigned. A proposed unseen-plant evaluation must keep all views of a plant together; no evaluation has occurred
Intended / excluded uses	Classroom documentation practice; no demonstrated outdoor use or disease diagnosis

Card field	Evidence-linked Photo practice v1 entry
Open questions	Exact label rule/records, source detail, permission for any proposed external reuse, quality checks, collection dates and assignment record

Explain the documentation mechanism

Read each entry as claim → evidence → status. A plan is not an observed result: “split by plant” is a proposed rule until exact assignments exist. An empty field is not evidence of absence: “labels not supplied” differs from “every photo has no spot.” Dataset-card fields are a classroom selection, not a claim to implement every research datasheet question.

No Python execution is required to establish the supplied18/6 counts; count the manifest groups and views directly.

▼ Review · Earlier trip-audit code and assumptions

```
rows = [("A", 8), ("B", None), ("B", None),
        ("C", -2), ("D", 12)]
ids = set()
missing = 0
negative = 0
for trip_id, minutes in rows:
    ids.add(trip_id)
    if minutes is None:
        missing = missing + 1
    elif minutes < 0:
        negative = negative + 1
print(len(rows), len(ids), missing, negative)
```

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

Turn an observation into a documented action

FINDING	WHAT IT ESTABLISHES	NEXT ACTION
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

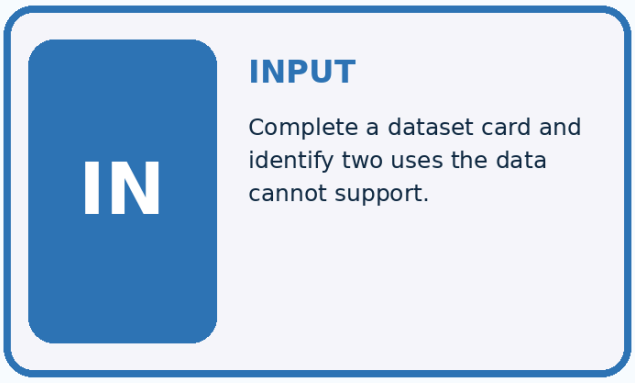
A successful run does not certify data quality.

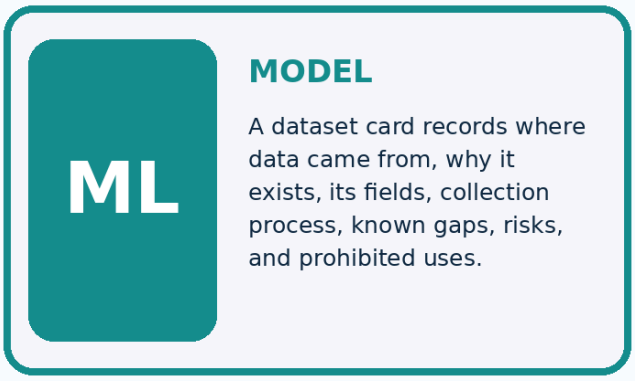
▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Complete a **dataset card** and identify two uses the data cannot support.

MODEL IDEA A **dataset card** records where data came from, why it exists, its fields, collection **process**, known gaps, risks, and **prohibited use**s.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated **limitation**.

UNDER THE HOOD Source: student-collected classroom plant photos with permission. Purpose: teach a low-stakes healthy-versus-spotted classifier.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

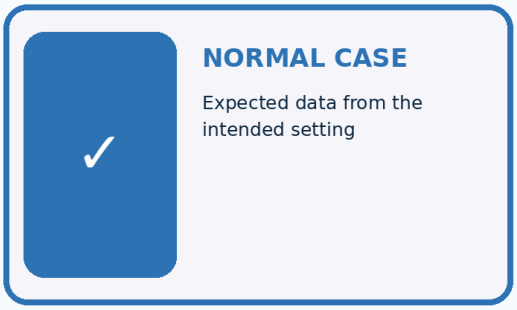
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

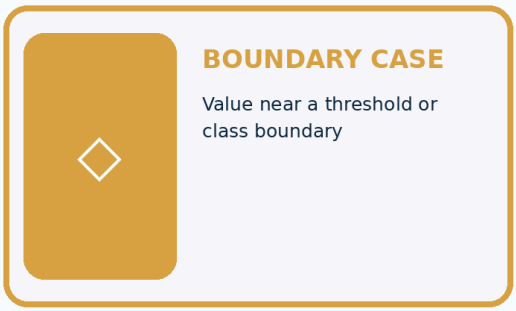
Compare an attractive photo count with a traceable dataset record

> Documentation reveals conditions and gaps; it does not repair them.

Four evaluation conditions



NORMAL Expected data and expected behavior.

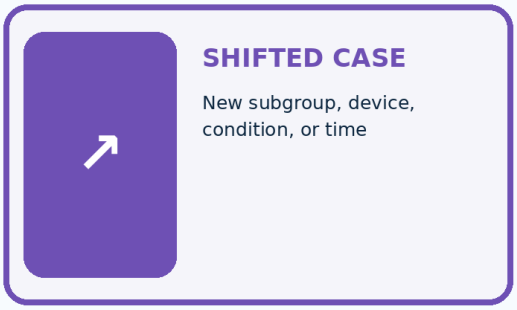


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The dataset adds 6 outdoor photos of the same plants.

▼ **Check after predicting**

Prediction: Photo count becomes 24, plant count remains 6; lighting coverage changes.

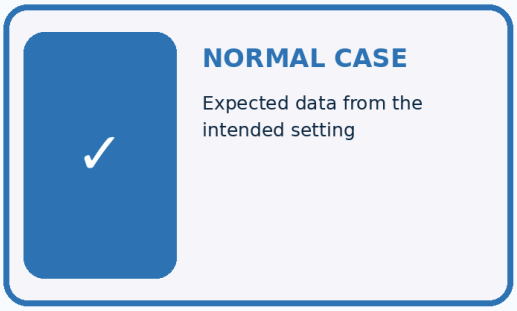
Reason: Additional views broaden conditions without creating new independent plant identities.

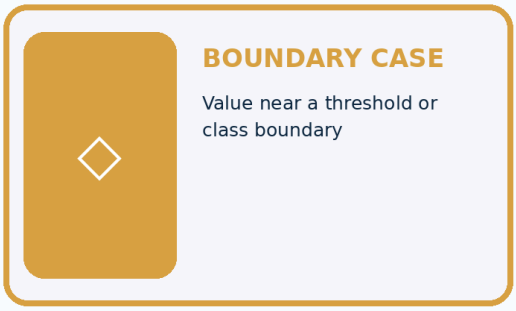
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

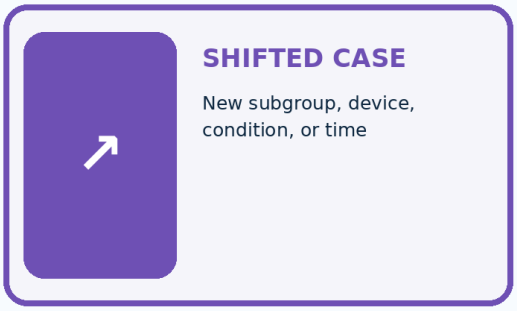
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Update the counts and conditions without changing the source identities

> Version the card and preserve the previous evidence.

Complete pictured facts and v2 update

Build a **dataset card** that reveals what the photo count hides.

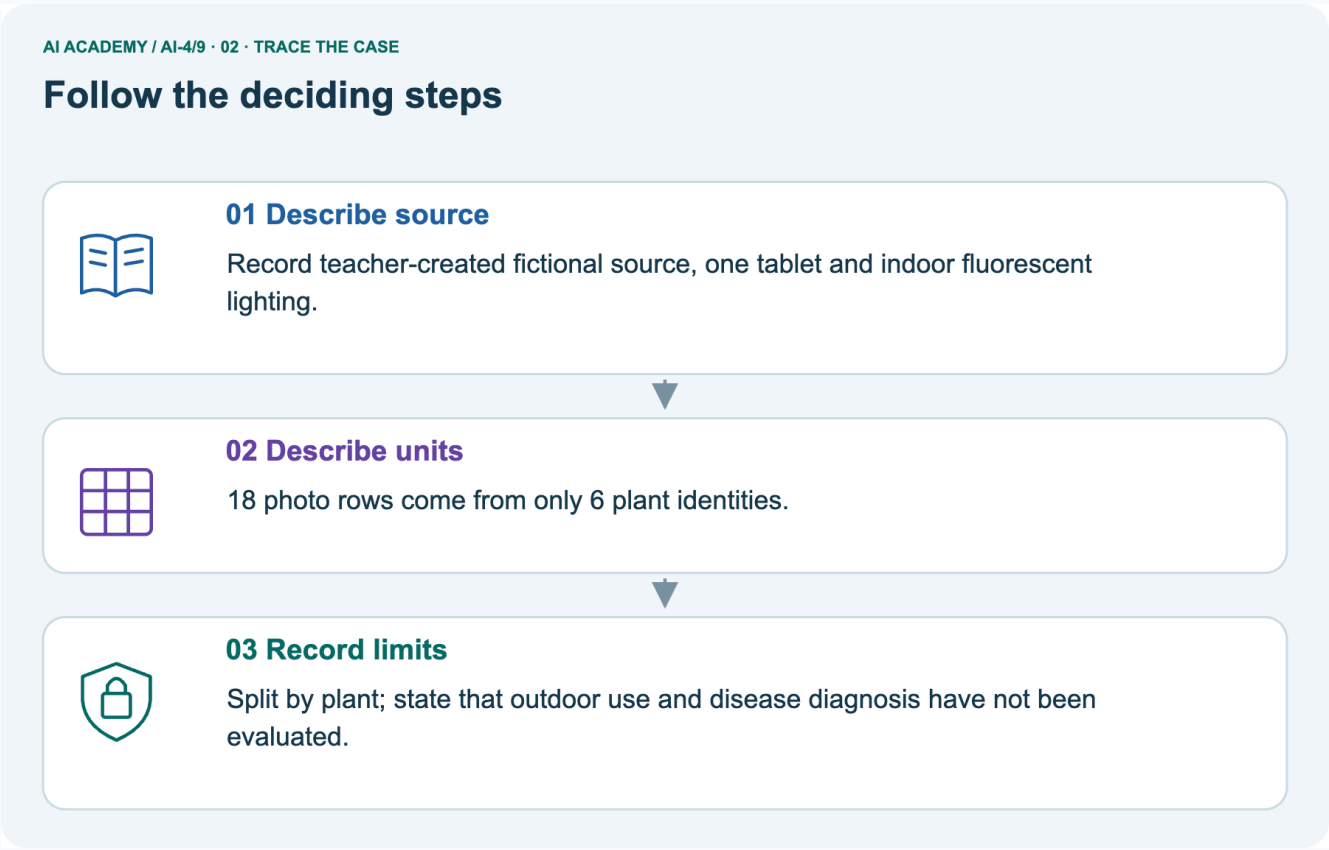
Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light. Label rule is visible spot/no visible spot/review.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Photo practice v2 adds one outdoor view of each of the same P1–P6 plants: new IDs P1d,P2d,P3d,P4d,P5d,P6d. The updated card records24 photos,6plant identities,18 indoor and6 outdoor views. Outdoor lighting details, device for the added views and per-photo labels are not supplied; do not invent them. A condition was added, but no additional source plants or accuracy result were created.

Preserve v1 and record this explicit change in v2.

Trace source, row units and limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Describe source	Record teacher-created fictional source, one tablet and indoor fluorescent lighting.
2. Describe units	18 photo rows come from only 6 plant identities.
3. Record limits	Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

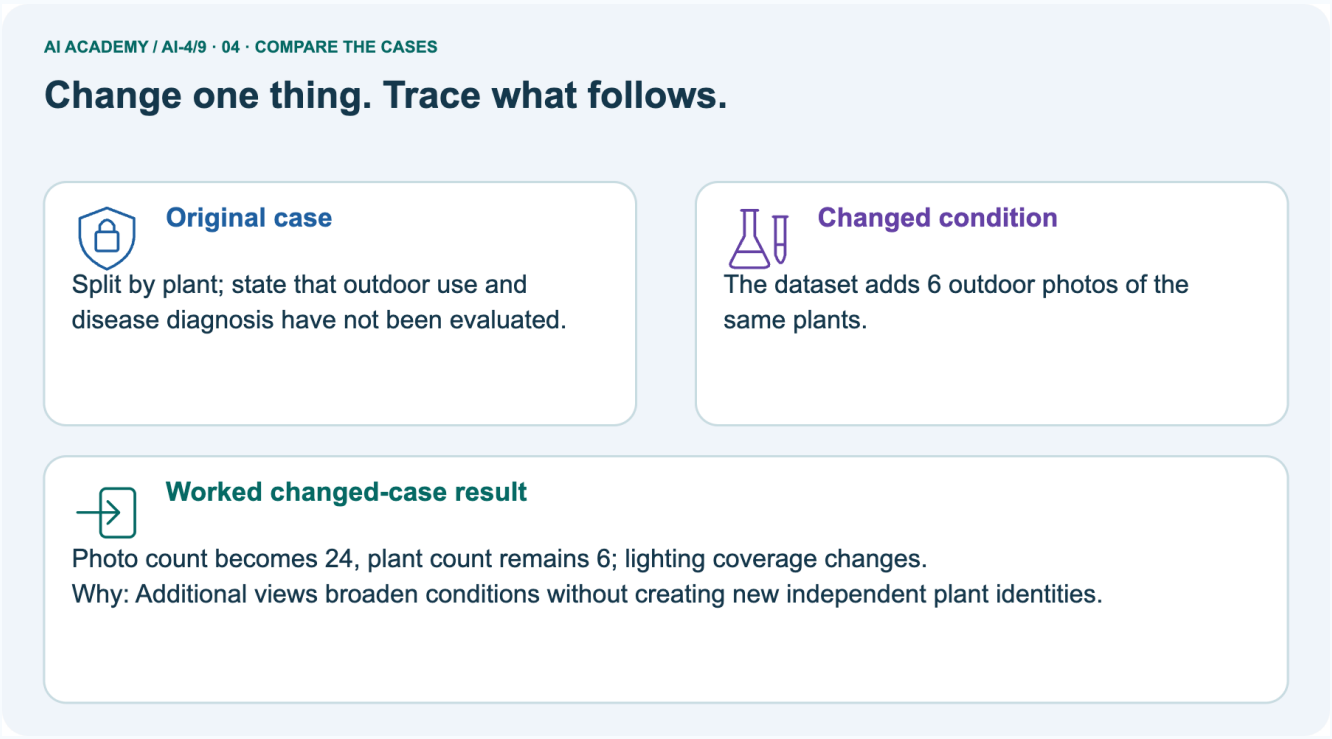
Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict six outdoor views of the same plants

- Change one thing: The dataset adds 6 outdoor photos of the same plants. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining gaps

- A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

- Read the labelled picture: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Earlier trip card and overlap question

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 9 • WORKED EVIDENCE

1. Source: student-collected classroom plant photos with permission.
2. Purpose: teach a low-stakes healthy-versus-spotted classifier.
3. Gap: few evening-light photos and no outdoor disease diagnosis evidence.
4. Prohibit medical, agricultural, or pesticide decisions; require teacher review.

STEP 1 Source: student-collected classroom plant photos with permission.

STEP 2 Purpose: teach a low-stakes healthy-versus-spotted classifier.

STEP 3 Gap: few evening-light photos and no outdoor disease diagnosis **evidence**.

STEP 4 Prohibit medical, agricultural, or pesticide decisions; require teacher review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

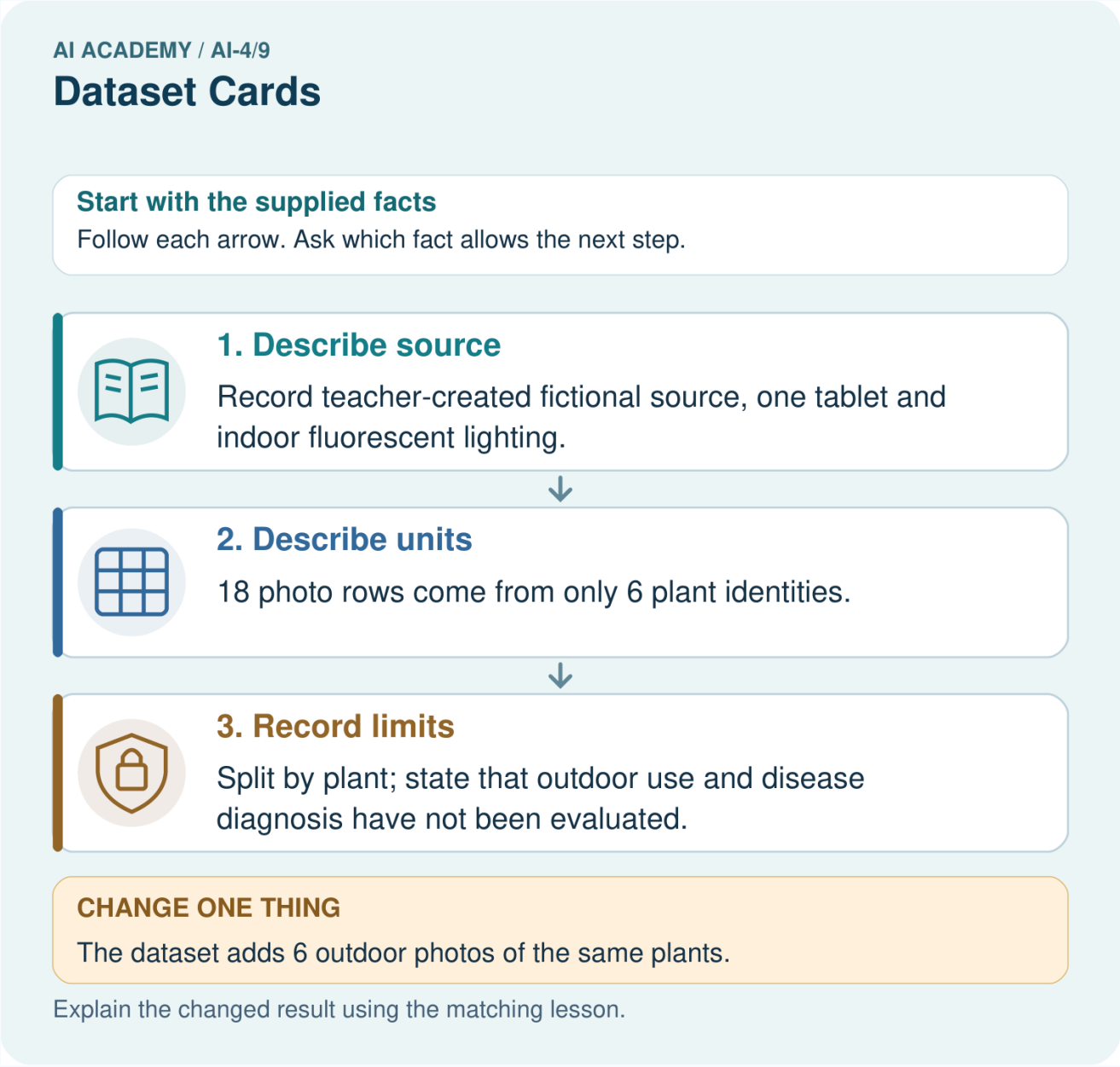
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dataset Cards

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find card entries that outrun their source

> Correct the claim and keep the unresolved question visible.

Complete misleading draft and repairs

Repair this draft: “18 images,18 independent plants, healthy labels, safe outdoors.” The manifest has6plant identities, per-photo labels are not supplied, and only indoor fluorescent conditions are described. The dataset card supports none of those added claims. A second draft says “the split is plant-grouped” when no assignments exist: record a proposed grouping rule and unassigned status instead.

Missing external reuse permission is a question to resolve for that use, not a value to guess.

Propose one evidence check

- Diagnose one cause and name the cleanest next experiment:

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A file name and list of columns are enough documentation for responsible reuse. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A file name and list of columns are enough documentation for responsible reuse.

A closer explanation

A complete-looking card does not prove the table is accurate or suitable for a model. It makes claims and unanswered questions reviewable.

Report raw counts separately from any later accepted subset, with explicit rules for missing, invalid and repeated records. Do not silently change denominators after cleaning.

For this exercise, zero is allowed and unknown is not zero. That is a stated **schema** choice, not a universal rule for all duration datasets.

In a real task, measurement resolution and event definitions determine which values make sense. Dataset inspection also does not replace protecting held out data from model choices.

A card says the dataset is ready because it has no empty cells. Name two checks that claim misses, then write a more accurate statement about Robot practice v1.

MAIN LESSON · SECTION 8

Practise writing and updating the card with less help

> Check every supplied fact against a card field.

Current card task and response

- From Photo practice v1's complete manifest, fill purpose/source/version, photo and plant counts, known conditions, label status and split status. Mark one field not supplied and one use with no supporting evaluation. Then update the card for v2's P1d–P6d outdoor additions, preserving the earlier version.

My dataset-card reasoning and evidence

Explain updated photo and source counts

- What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent five-plant card

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier trip-audit practice

A learner replaces both None values with 0 and removes the negative sign from -2. The table now passes a simple nonnegative check, but those edits invent durations.

Keep the original values, record the reason for each flag, and check the source. If a correction is supported, save a new version with the original value, corrected value and reason. If not, keep the uncertainty visible.

Another learner says all robot trips are covered because every row says indoor. A **dataset card** should instead state that outdoor trips are absent and that the data is fictional.

Extra copies of existing rows do not create new conditions or new independent **evidence**.

Use workbook W2 for the connected example “Dataset inspection and cards”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

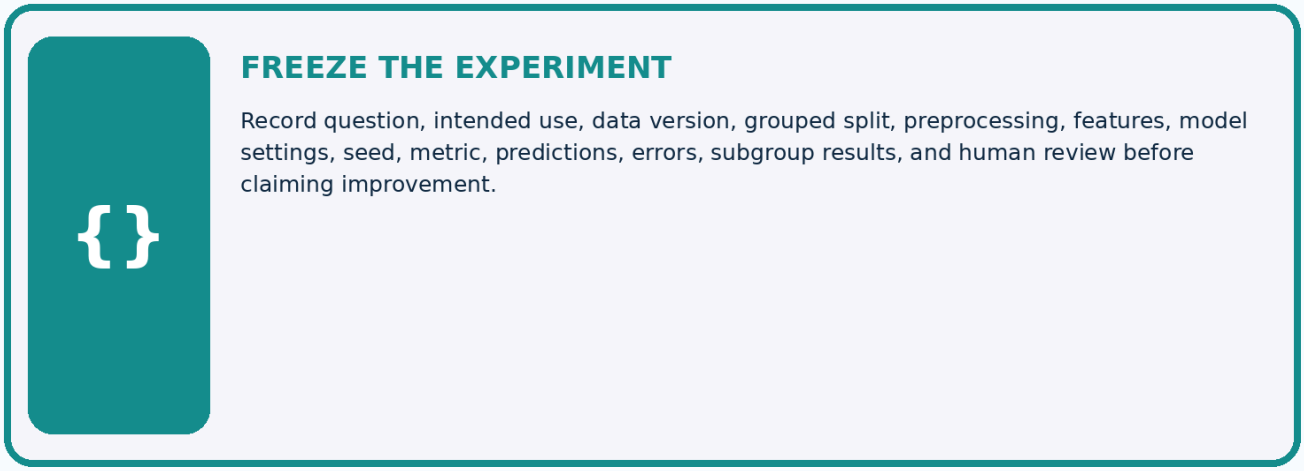
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + **limitation**s

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Interpret the audit counts

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete card-writing and update review

> Keep evidence, unknowns and future plans in distinct sentences.

Normal, source-boundary, missing-label and changed-version probes

Complete Card Review probe	Expected evidence decision
Normal:18photos from6plants/3views each	Record photo rows and source groups separately.
Boundary: record source is teacher-created fictional, not real collected data	State the fictional source; do not invent permission from real participants or field collection.
Missing: no per-photo label assignments supplied	Record vocabulary plus missing label records; no class counts or accuracy.
Changed source: v2 addsP1d–P6d outdoor views	24photos/6plants; conditions broaden, identities unchanged; unknown added-view details remain.

Write, trace, repair and version

1. Use the complete v1 manifest and worked card as the shared source. First write your own card from the facts, then compare every claim with the supplied evidence. Mark source, counts, conditions, labels, changes, split status, owner, excluded uses and unanswered questions. Repair the misleading draft without inventing replacements. Apply the exact v2 change in a new version; retain the old card.

   No model is fitted or accuracy measured. Keep the selected20-photo/5plant task for a first independent attempt.

Retain the evidence record

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

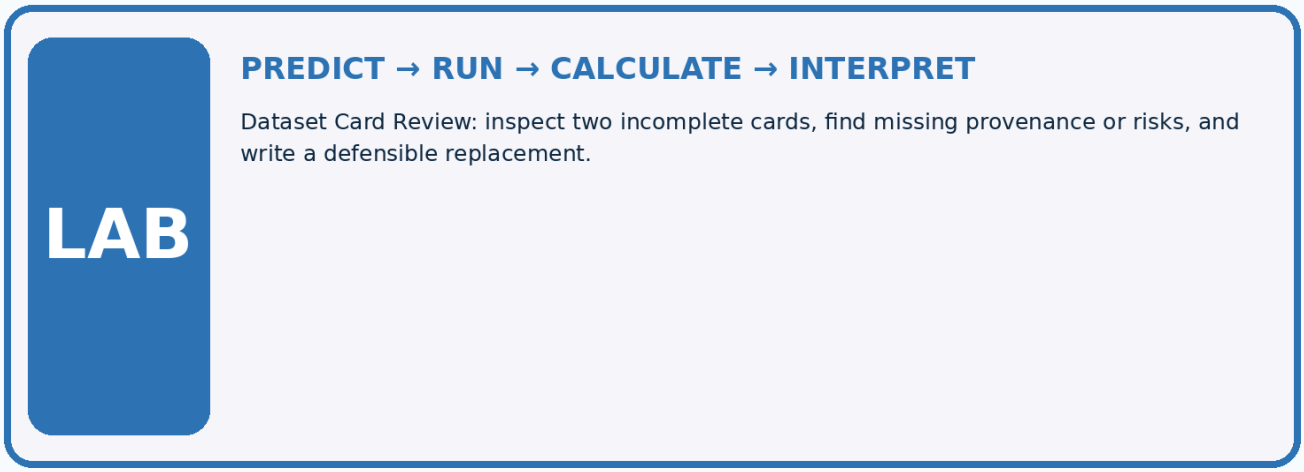
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. **Dataset Card** Review: inspect two incomplete cards, find missing **provenance** or risks, and write a defensible replacement.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: document a fresh twenty-photo dataset

> Make unknown fields visible instead of filling them with plausible guesses.

Selected facts and scope

Write key dataset-card entries for20 photos: five plants, four views each, one phone, one indoor setting. Growth labels have not been supplied.

Only the listed20photos/5plants/4views each,one phone,one indoor setting and absent growth labels are supplied. Do not invent the creator, date, permission, species, class balance or growth outcomes. Record unknown source fields as questions, propose a label contract and plant grouping, and distinguish proposals from completed assignments or evaluation.

A claim about growth needs a defined target and observations, not just photos.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Changed independent case

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A **dataset card** lists 500 images but omits collection conditions and intended use. Add two essential entries and explain their value.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Publish the Plant Classifier **dataset card** before model experiments begin; update it when data changes.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the project card ready for the baseline lesson

> A later experiment must name the exact dataset version it uses.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Publish the Plant Classifier **dataset card** before model experiments begin; update it when data changes.

Prepare the project dataset card for the classroom repository: source/version/owner, intended and excluded uses, row and group counts, collection conditions, label rule/history, quality checks, transformations, split assignments or explicit unassigned status, and remaining questions. Preserve links to raw evidence and updates.

A card does not by itself authorize external reuse or certify model performance; resolve those questions with the relevant evidence before that use.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

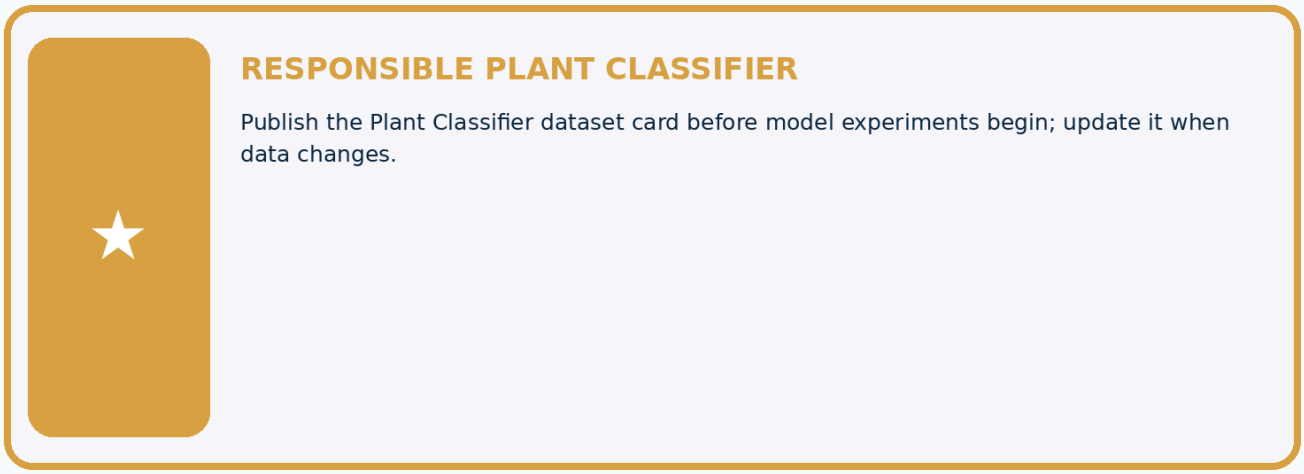
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with one supported entry and one open question

> Explain why a complete-looking card cannot certify quality.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Inspect row meaning **missing value**s and repeated records

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
dataset card	structured documentation for a dataset	
provenance	where data originated and changed	
limitation	condition weakening valid use	
prohibited use	use outside evidence or permission	

Closer-explanation terms

Word	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero

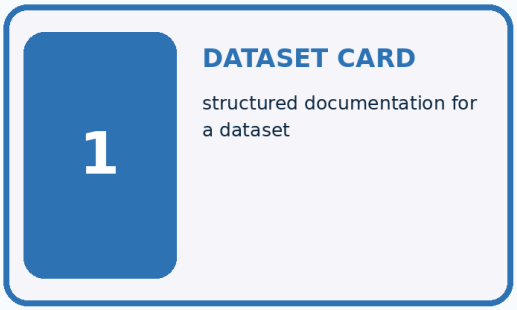
Word	Meaning
Dataset card	A concise record of dataset contents creation intended use and limitations

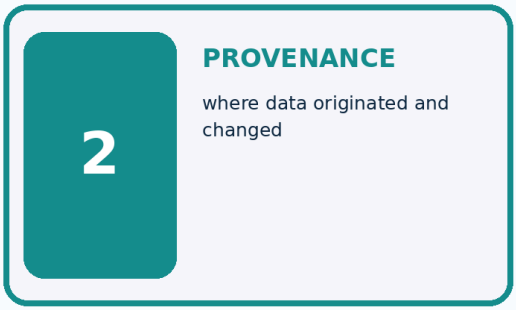
▼ Earlier wording, alternatives and picture transcripts

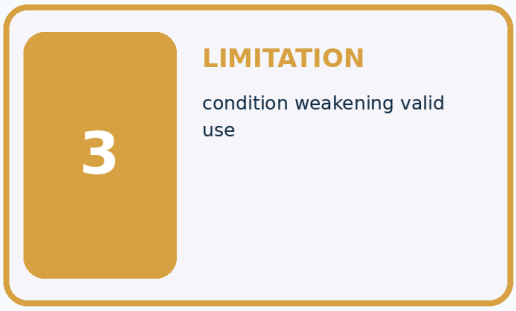
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A file name and list of columns are enough documentation for responsible reuse.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **dataset card**, **provenance**, **limitation**, **prohibited use**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A file name and list of columns are enough documentation for responsible reuse.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, **limitation**, and human responsibility.

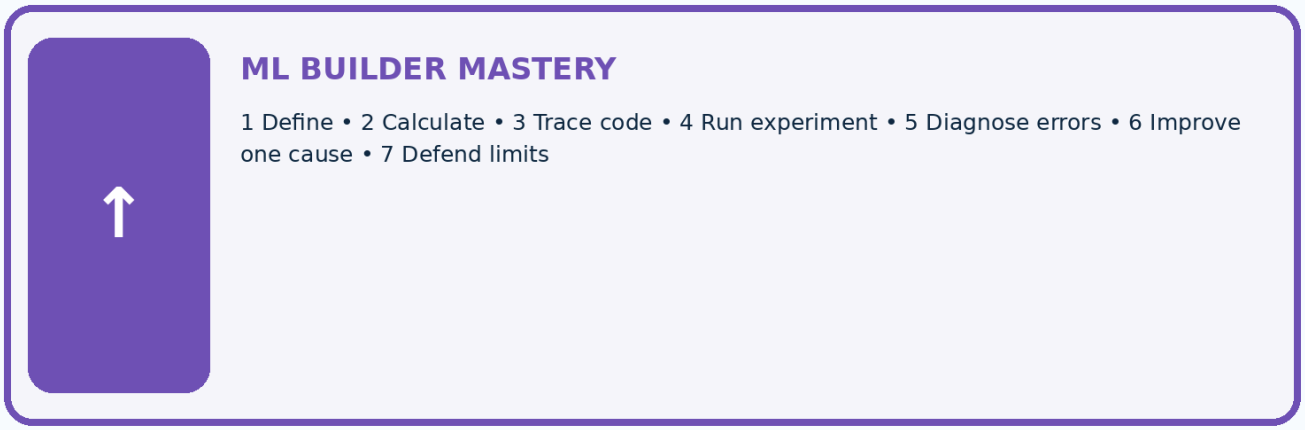
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected20-photo five-plant documentation task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a **limitation**, fairness concern, and the human-review action.

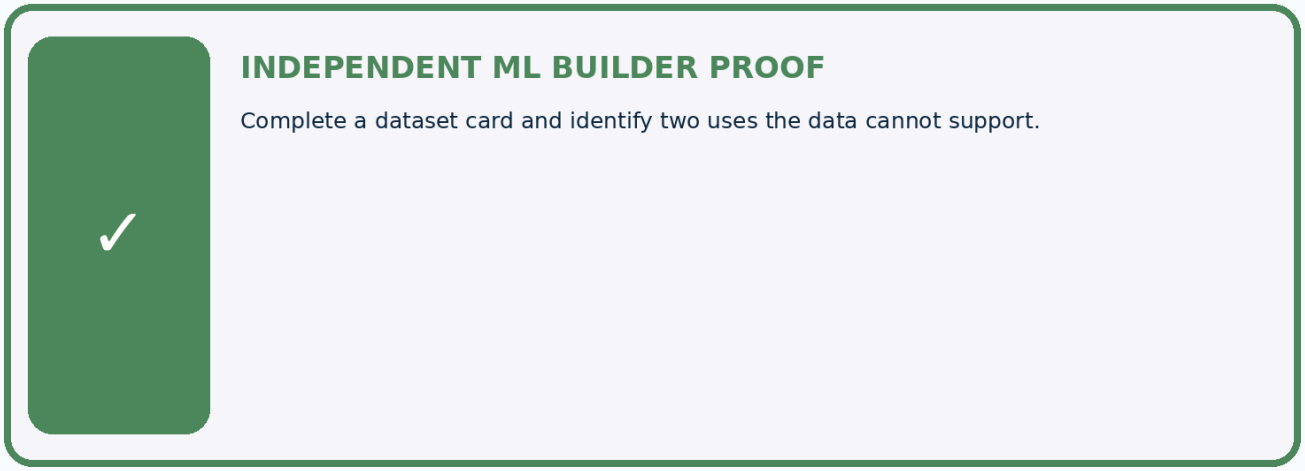
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Complete a **dataset card** and identify two uses the data cannot support.

NEXT BRIDGE Carry the experiment **evidence** into: Baselines First.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A **dataset card** lists fields but omits source and intended use.

Add information needed to assess suitability.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Provenance, permissions, collection method/time/conditions, label definitions, coverage, exclusions, missingness, use and **limitation**s. Fields alone are insufficient.

Evidence or stage	Value or result
Origin	Source, permission and collection
Meaning	Fields and operational labels
Coverage	Conditions, exclusions and missingness
Use	Intended tasks and limitations

▼ Earlier wording, alternatives and picture transcripts

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A dataset card explains suitability

A **dataset card** lists fields but omits source and intended use.

Add information needed to assess suitability.

Provenance, permissions, collection method/time/conditions, label definitions, coverage, exclusions, missingness, use and **limitation**s. Fields alone are insufficient.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Inspect row meaning missing values and repeated records	Explain the deciding step, show a trace or test, and state one limit.
Document dataset source schema and limitations	Explain the deciding step, show a trace or test, and state one limit.
Separate observed quality flags from justified corrections	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Build a **dataset card** that reveals what the photo count hides.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Build a **dataset card** that reveals what the photo count hides.

- Which labelled step matches this detail? Split by plant; state that outdoor use and disease diagnosis have not been evaluated.
- 1. Record limits
- 2. Describe units
- 3. Describe source
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Record limits Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Read the picture in order.

Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The dataset adds 6 outdoor photos of the same plants. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Photo count becomes 24, plant count remains 6; lighting coverage changes.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Photo count becomes 24, plant count remains 6; lighting coverage changes. Additional views broaden conditions without creating new independent plant identities.

Repair and keep the record

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · The six-picture journey</summary>

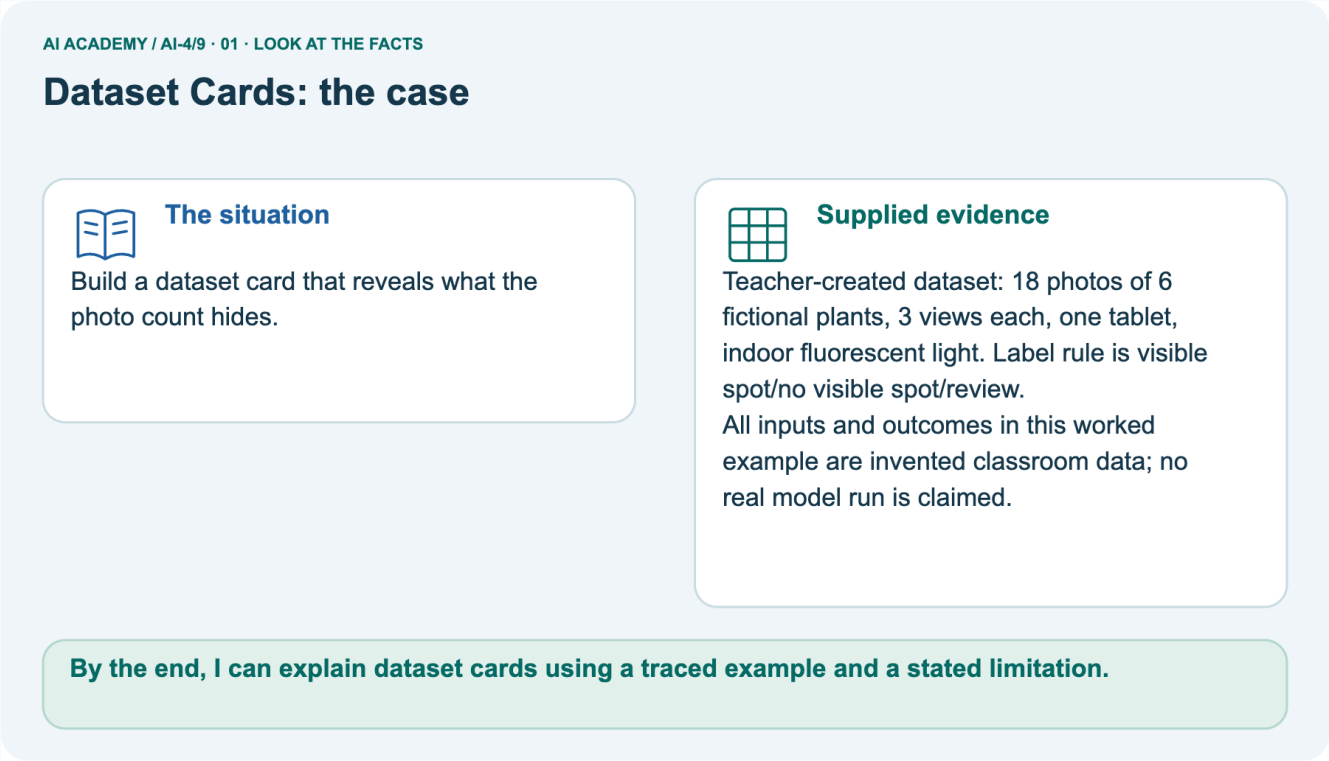
</details>

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Dataset Cards: the case



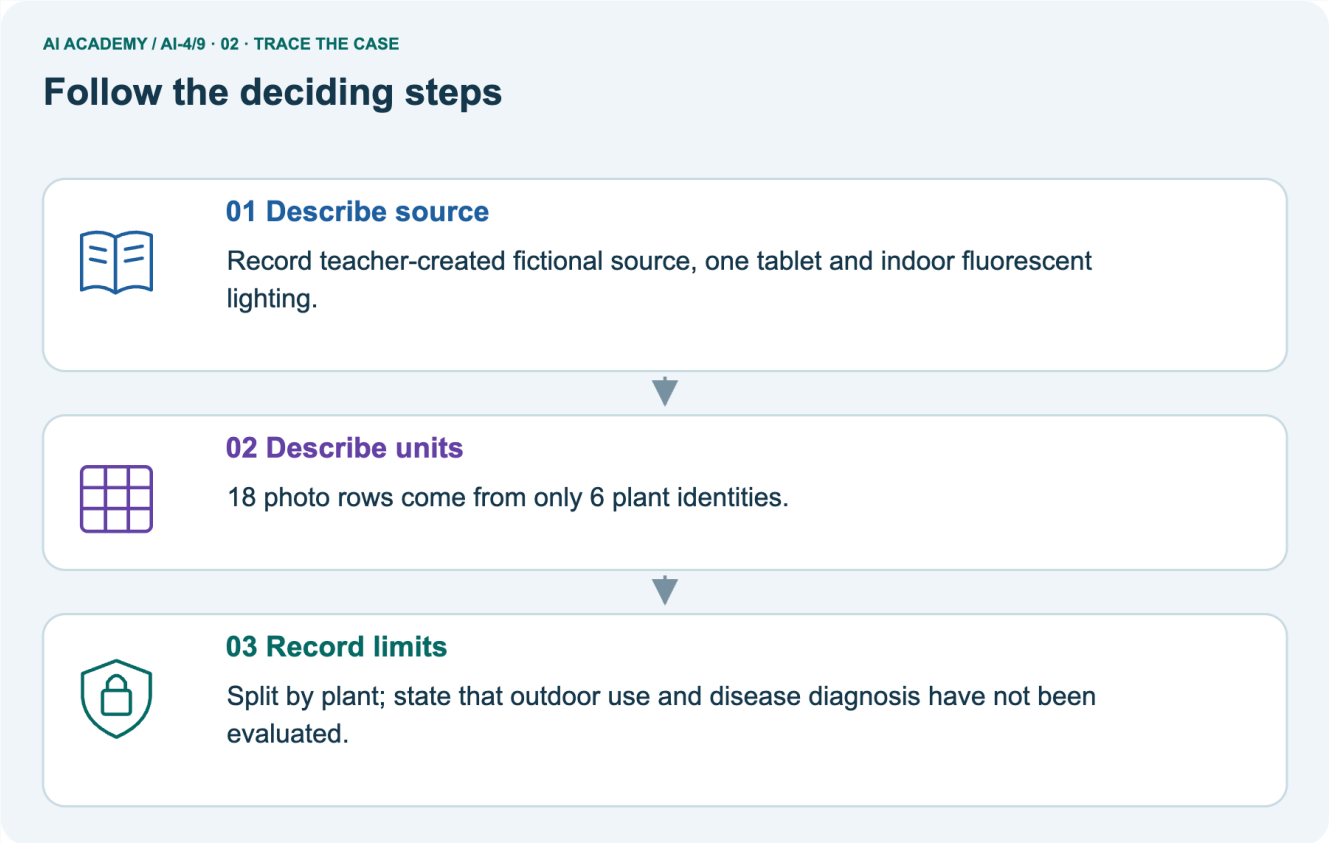
Follow the labelled picture with the matching explanation. Open the vector diagram

Build a **dataset card** that reveals what the photo count hides. Supplied facts: Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light.

Label rule is visible spot/no visible spot/review. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

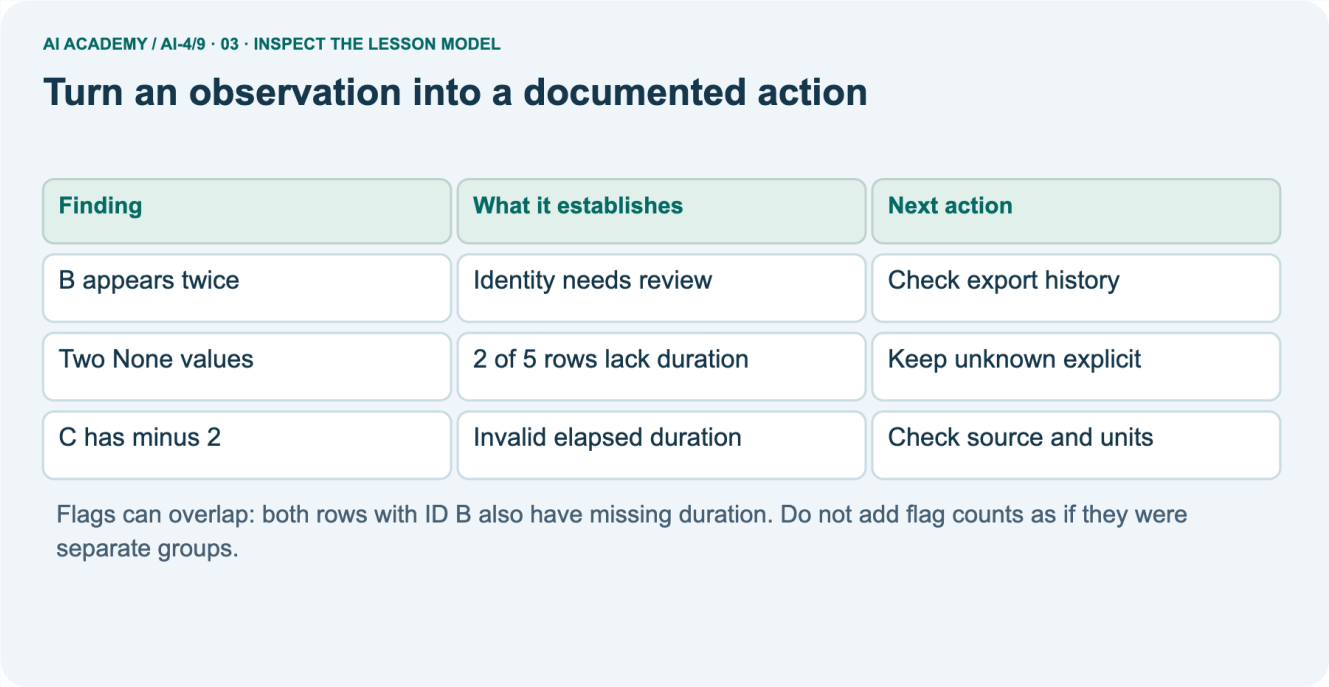


Follow the labelled picture with the matching explanation. Open the vector diagram

Describe source: Record teacher-created fictional source, one tablet and indoor fluorescent lighting. → Describe units: 18 photo rows come from only 6 plant identities. → Record limits: Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Point to each numbered stage and explain which recorded **input** it uses.

03. Turn an observation into a documented action



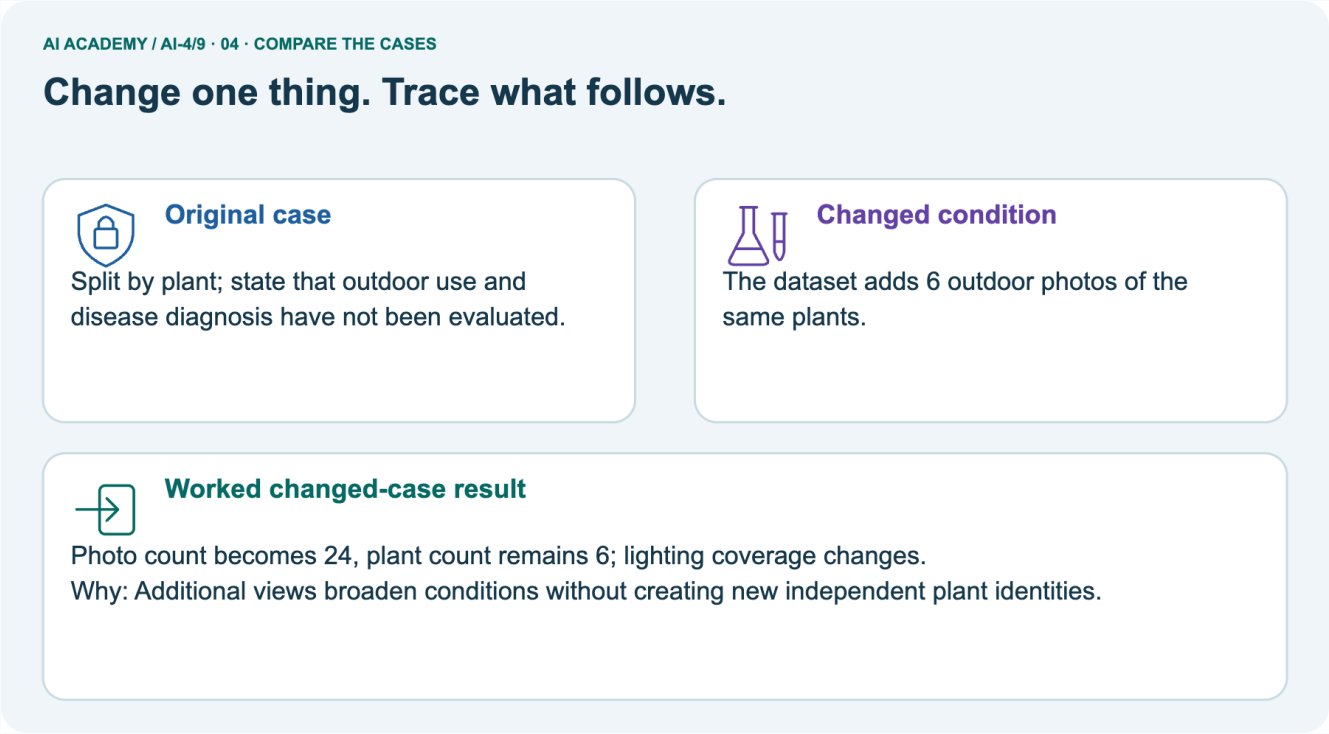
Follow the labelled picture with the matching explanation. Open the vector diagram

Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



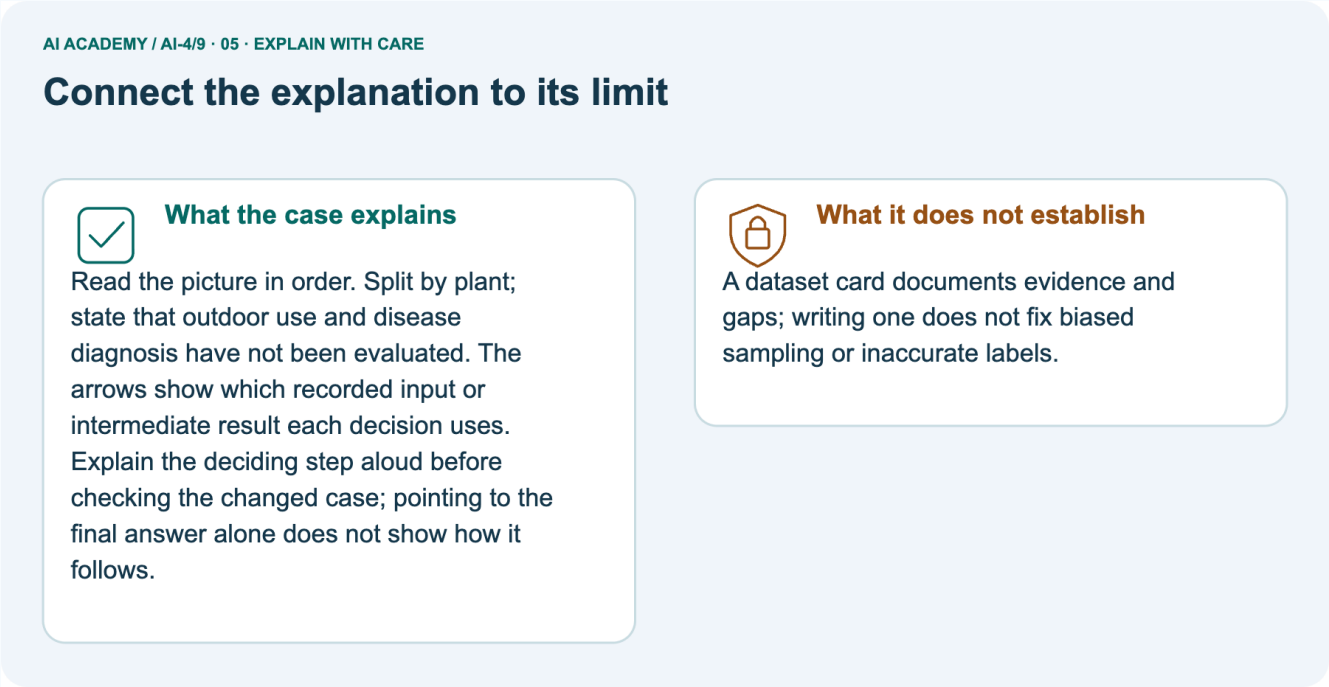
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Changed condition: The dataset adds 6 outdoor photos of the same plants.

Expected result: Photo count becomes 24, plant count remains 6; lighting coverage changes. Reason: Additional views broaden conditions without creating new independent plant identities.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

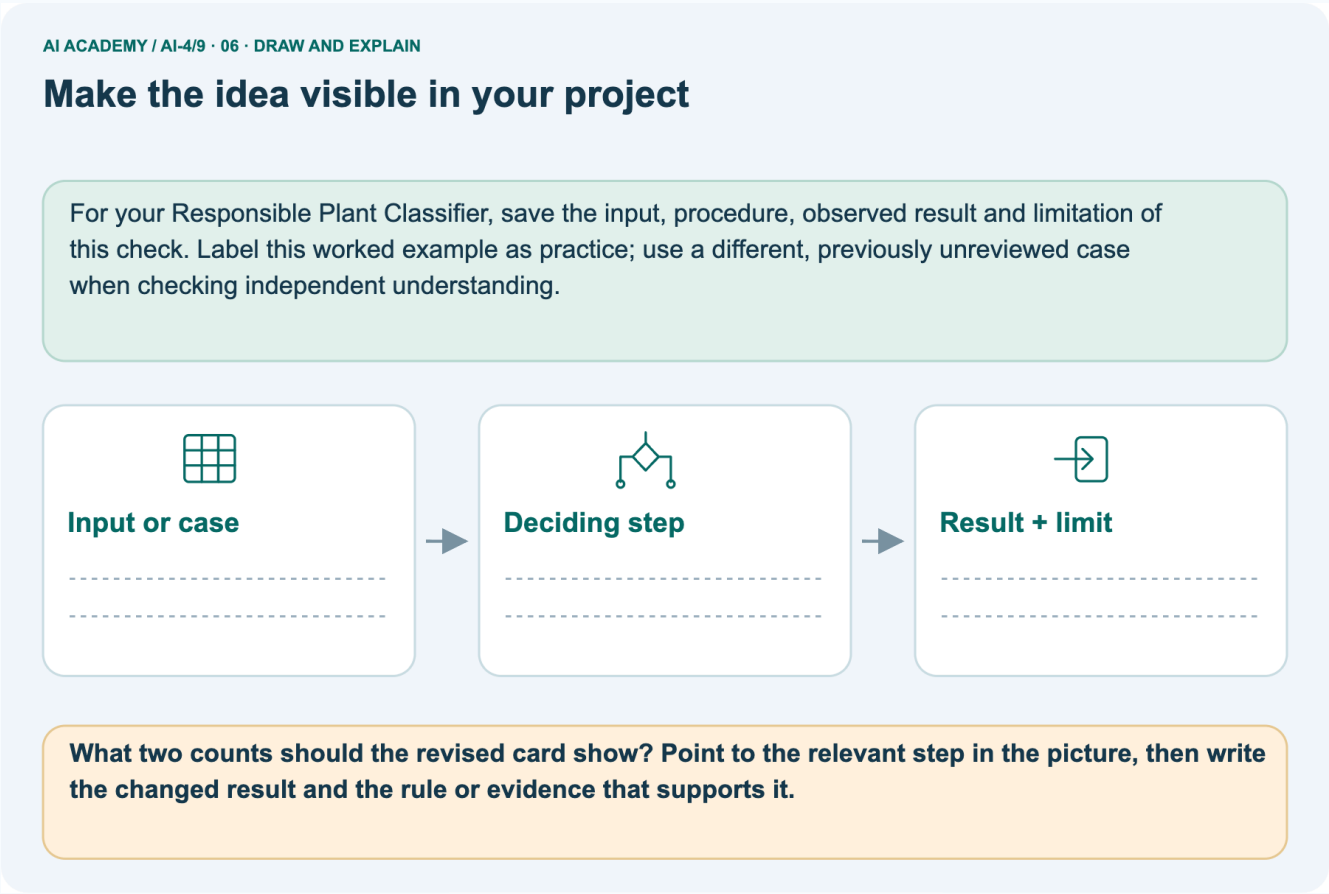
Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Dataset Cards: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.